Symposium on the Theory of Models; North-Holland Publ. Co., Amsterdam (1965)

REDUCTIONS OF HIGHER-ORDER LOGIC *

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A higher-order logic is considered which contains variables of all finite and transfinite types, together with a more restricted logic in which the types of variables are all in a sense describable ordinals. It is shown that several important problems connected with the restricted higher-order logic, pertaining, for instance, to the determination of spectra and the characterization of logical truth, are reducible to the corresponding problems for second-order logic; the principal results of this sort extend earlier results of Zykov and Hintikka. It is obtained as a corollary that the set of logical truths of second-order logic does not appear in any reasonable extension of the Kleene hierarchy.

1. Preliminaries. As ingredients of *second-order formulas* we assume the following disjoint categories of *symbols* to be available: (1) the usual sentential connectives $\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, and brackets, (2) a denumerable set of *individual variables*, (3) for each positive integer n, a denumerable set of n-place *predicate variables*, (4) the two quantifiers $\bigwedge$ and $\bigvee$ (respectively universal and existential), (5) for each natural number n, a proper class [1] of n-place *predicate constants*.

Each symbol is assumed to be a 1-place sequence. By a *second-order expression* is understood a finite sequence each of whose 1-place subse-

* The preparation of most of this paper was supported by the United States National Science Foundation, under grant NSF–GP–1603 (Montague). I am indebted for helpful comments to Professor Robert L. Vaught and my students Dr. David Kaplan and Mr. Barry Kurtzman.

[1] As a basis for the metatheory is assumed a system of set theory involving both sets (which are members of something) and proper classes (which are members of nothing); a class is either a set or a proper class. To be explicit, I have in mind the axomiatic theory of Bernays–Morse, including the Axiom of Choice, as presented in the Appendix of Kelley [55]. The *ordinals* are defined in such a way that each ordinal is the set of all smaller ordinals, and the *cardinal* of a set A is the least ordinal with the same cardinality as A. Thus the cardinals form a subclass of the ordinals.

quences is a symbol of one of the sorts listed above. The class of *second-order formulas* is the least class Φ such that (1) Φ contains all second-order expressions $\mathsf{P}\mathsf{u}_0 \ldots \mathsf{u}_{n-1}$, where n is a positive integer, P is an n-place predicate constant or predicate variable, and $\mathsf{u}_0, \ldots, \mathsf{u}_{n-1}$ are individual variables, (2) Φ is closed under the application of sentential connectives, and (3) $\bigwedge \mathsf{u} \phi$ and $\bigvee \mathsf{u} \phi$ are in Φ whenever ϕ is in Φ and u is either an individual variable or a predicate variable.

We shall employ the usual notion of a model with one modification: models with an empty universe will be admitted. Thus a *model* is to be a pair $\langle A, F \rangle$ such that A is a set (possibly empty), F is a function whose domain is a set of predicate constants, and whenever P is an n-place predicate constant in the domain of F, $F(\mathsf{P})$ is an n-place relation on A (that is, a set of ordered n-tuples of members of A).

If $\mathfrak{A} = \langle A, F \rangle$, then we understand by the *universe* of $\mathfrak{A}$, or $U_{\mathfrak{A}}$, the set A, and by the *language* of $\mathfrak{A}$ the domain of F; if P is in the language of $\mathfrak{A}$, then $\mathsf{P}_{\mathfrak{A}}$ is to be $F(\mathsf{P})$. If $\mathfrak{A}$, $\mathfrak{B}$ are models, then $\mathfrak{A}$ is said to be *isomorphic* to $\mathfrak{B}$ or a *submodel* of $\mathfrak{B}$ under the usual conditions (including the requirement that $\mathfrak{A}$ and $\mathfrak{B}$ have the same language). By an *L-model* is understood a model whose language is L. With each model $\mathfrak{A}$ we can associate an *isomorphism type* $\tau\mathfrak{A}$ in such a way that (1) $\tau\mathfrak{A}$ is always a set, and (2) for any models $\mathfrak{A}$ and $\mathfrak{B}$ having the same language, $\tau\mathfrak{A} = \tau\mathfrak{B}$ if and only if $\mathfrak{A}$ is isomorphic to $\mathfrak{B}$ [2]. If L is a set of predicate constants, then by an *L-type* we understand an isomorphism type α such that $\alpha = \tau\mathfrak{A}$ for some L-model $\mathfrak{A}$. If α, β are isomorphism types, we say that $\alpha \leqslant \beta$ if and only if there are models $\mathfrak{A}$, $\mathfrak{B}$ such that $\tau\mathfrak{A} = \alpha$, $\tau\mathfrak{B} = \beta$, and $\mathfrak{A}$ is a submodel of $\mathfrak{B}$.

Definition 1. *If ϕ is a second-order formula whose free variables are all individual variables and in the standard ordering are $\mathsf{v}_0, \ldots, \mathsf{v}_{n-1}$, then ϕ is said to be* satisfied *in a model $\mathfrak{A}$ by a sequence $\langle a_0, \ldots, a_{n-1} \rangle$ of elements of $U_{\mathfrak{A}}$ if* (1) *the predicate constants occurring in ϕ are all in the language of $\mathfrak{A}$, and* (2) *ϕ is true when* (i) *each predicate constant P occurring in ϕ is regarded as denoting $\mathsf{P}_{\mathfrak{A}}$,* (ii) *the variables $\mathsf{v}_0, \ldots, \mathsf{v}_{n-1}$ are regarded (when occurring freely) as denoting $a_0, \ldots, a_{n-1}$ respectively,* (iii) *the bound*

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[2] We could for instance define $\tau\mathfrak{A}$ as the set of all models $\mathfrak{B}$ such that $\mathfrak{A}$ is isomorphic to $\mathfrak{B}$ and $U_{\mathfrak{B}}$ is the cardinal of $U_{\mathfrak{A}}$. I prefer not to select at this point any particular definition of isomorphism type but to employ only the properties (1) and (2). This will permit the later convenience of identifying particular kinds of isomorphism types with particular set-theoretical objects in an *ad hoc* manner.

individual variables of ϕ *are regarded as ranging over* $U_{\mathfrak{A}}$, *and* (iv) *for each positive integer* n, *the bound* n*-place predicate variables of* ϕ *are regarded as ranging over the set of all* n*-place relations on* $U_{\mathfrak{A}}$. (In this definition, as in others below, an intuitive notion of truth is used, but one which can be exactly defined by well-known methods, like those in Tarski and Vaught [57]. Slight modifications, required to accommodate empty models, are spelled out in Kalish and Montague [a].) *We then say that* ϕ *is* true *in* $\mathfrak{A}$ *if* ϕ *contains no free variables and* ϕ *is satisfied in* $\mathfrak{A}$ *by the empty sequence* Λ.

As ingredients of *higher-order formulas* we assume the following disjoint categories of *symbols* to be available: (1) the sentential connectives, brackets, quantifiers, and predicate constants introduced above, (2) for each ordinal α, a denumerable set of *variables of type* α (the variables of types 0 and 1 coinciding respectively with our former individual variables and 1-place predicate variables), and (3) the symbol η, regarded as indicating membership, and the identity symbol $=$.

As before, each of these symbols is assumed to be a 1-place sequence, and a *higher-order expression* is to be a finite sequence each of whose 1-place subsequences is one of these symbols. The class of *higher-order formulas* is the least class Φ such that (1) Φ contains all higher-order expressions $\mathsf{u}\eta\mathsf{v}$, $\mathsf{u}=\mathsf{v}$, $\mathsf{P}\mathsf{u}_0 \ldots \mathsf{u}_{n-1}$, where u, v are variables of arbitrary type, n is a positive integer, P is an n-place predicate constant, and $\mathsf{u}_0, \ldots, \mathsf{u}_{n-1}$ are variables of type 0, (2) Φ is closed under the application of sentential connectives, and (3) $\bigwedge\mathsf{u}\phi$ and $\bigvee\mathsf{u}\phi$ are in Φ whenever ϕ is in Φ and u is a variable of any type.

We shall now extend the notions of satisfaction and truth to arbitrary higher-order formulas. A consideration of transfinite types suggests departing from a direct generalization of Definition 1; contrary to the course exemplified in clause (iv) of that definition, we shall regard the types as cumulative. There are several reasons for considering the notion introduced below to be the correct notion of truth for higher-order formulas; but we shall proceed rather arbitrarily here, leaving a detailed justification of the definition to the paper Montague [a].

Let ε be a 2-place predicate constant and Σ a 1-place predicate constant; we regard these symbols as expressing membership and the property of being a class (or non-individual), respectively. Let $\mathsf{t}, \ldots, \mathsf{z}$ be distinct individual variables and P a 1-place predicate variable. We understand by $\mathrm{RFS}_{\varepsilon, \Sigma, \mathsf{t}, \ldots, \mathsf{z}, \mathsf{P}}$ the universal closure of the conjunction of the following formulas (which are given in Montague–Scott–Tarski [a] as the axioms for *second-order rank-free set theory with individuals*; both here and below ε is for perspicuity written between its two arguments rather than before them, as strict conformity to our definitions would require):

$$\Lambda \mathsf{x}\,[\mathsf{x}\varepsilon\mathsf{u} \leftrightarrow \mathsf{x}\varepsilon\mathsf{v}] \wedge \Sigma\mathsf{u} \wedge \Sigma\mathsf{v} \rightarrow \Lambda\mathsf{P}\,[\mathsf{Pu} \leftrightarrow \mathsf{Pv}],$$

$$\Sigma\mathsf{u} \rightarrow \Lambda\mathsf{P}\mathsf{V}\mathsf{v}\,[\Sigma\mathsf{v} \wedge \Lambda\mathsf{x}\,[\mathsf{x}\varepsilon\mathsf{v} \leftrightarrow \mathsf{x}\varepsilon\mathsf{u} \wedge \mathsf{Px}]],$$

$$\Sigma\mathsf{u} \rightarrow \mathsf{V}\mathsf{w}\Lambda\mathsf{y}\,[\mathsf{y}\varepsilon\mathsf{w} \leftrightarrow \neg\,\Sigma\mathsf{y} \vee \mathsf{V}\mathsf{v}(\mathsf{v}\varepsilon\mathsf{u} \wedge \Lambda\mathsf{t}\,[\mathsf{t}\varepsilon\mathsf{y} \rightarrow \mathsf{t}\varepsilon\mathsf{v}])],$$

$$\Lambda\mathsf{x}\mathsf{V}\mathsf{z}\Lambda\mathsf{u}\Lambda\mathsf{w}\,[\Lambda\mathsf{y}\,[\mathsf{y}\varepsilon\mathsf{u} \rightarrow \mathsf{y}\varepsilon\mathsf{z}] \wedge \Lambda\mathsf{y}\,[\mathsf{y}\varepsilon\mathsf{w} \leftrightarrow \neg\,\Sigma\mathsf{y} \vee$$
$$\mathsf{V}\mathsf{v}(\mathsf{v}\varepsilon\mathsf{u} \wedge \Lambda\mathsf{t}\,[\mathsf{t}\varepsilon\mathsf{y} \rightarrow \mathsf{t}\varepsilon\mathsf{v}])] \rightarrow \mathsf{w}\varepsilon\mathsf{z} \vee$$
$$\Lambda\mathsf{y}\,[\mathsf{y}\varepsilon\mathsf{x} \rightarrow \mathsf{y}\varepsilon\mathsf{w}]],$$

$$\mathsf{y}\varepsilon\mathsf{x} \rightarrow \Sigma\mathsf{x}.$$

Let $\mathfrak{A}$ be an $\{\varepsilon, \Sigma\}$-model in which the second-order formula $\mathrm{RFS}_{\varepsilon,\Sigma,\mathsf{t},\ldots,\mathsf{z},\mathsf{P}}$ is true. Then $T_{\mathfrak{A}}(0)$ is to be the set of individuals of $\mathfrak{A}$; more generally, if α is any ordinal, $T_{\mathfrak{A}}(\alpha)$ is to be the αth cumulative type in $\mathfrak{A}$, in accordance with the following recursive definition. (1) $T_{\mathfrak{A}}(0)$ is the set of objects x in $U_{\mathfrak{A}}$ such that $\langle x\rangle$ is not in $\Sigma_{\mathfrak{A}}$. (2) $T_{\mathfrak{A}}(\alpha+1)$ is the set of objects x in $U_{\mathfrak{A}}$ such that for all y, if $\langle y, x\rangle$ is in $\varepsilon_{\mathfrak{A}}$, then y is in $T_{\mathfrak{A}}(\alpha)$. (3) if λ is a limit number, then $T_{\mathfrak{A}}(\lambda)$ is the union of the sets $T_{\mathfrak{A}}(\alpha)$ for $\alpha<\lambda$. It turns out that $U_{\mathfrak{A}}$ is $T_{\mathfrak{A}}(\alpha)$ for some ordinal α; we call the least such α the *rank* of $\mathfrak{A}$.

Definition 2. *If ϕ is a higher-order formula whose free variables are all individual variables and are in the standard ordering $\mathsf{v}_0, \ldots, \mathsf{v}_{n-1}$, then we say that ϕ is* satisfied *in a model $\mathfrak{A}$ by a sequence $\langle a_0, \ldots, a_{n-1}\rangle$ of elements of $U_{\mathfrak{A}}$ if* (1) *the predicate constants occurring in ϕ are all in the language of $\mathfrak{A}$, and there exist ε, Σ, $\mathsf{t}, \ldots, \mathsf{z}$, P, $\mathfrak{B}$ such that* (2) *ε, Σ are respectively a 2-place and a 1-place predicate constant,* (3) *$\mathsf{t}, \ldots, \mathsf{z}$ are distinct individual variables,* (4) *P is a 1-place predicate variable,* (5) *$\mathfrak{B}$ is an $\{\varepsilon, \Sigma\}$-model in which $\mathrm{RFS}_{\varepsilon,\Sigma,\mathsf{t},\ldots,\mathsf{z},\mathsf{P}}$ is true,* (6) *$T_{\mathfrak{B}}(0)$ is $U_{\mathfrak{A}}$,* (7) *all variables occurring in ϕ have types which are at most the rank of $\mathfrak{B}$, and* (8) *ϕ is true when* (i) *each predicate constant P occurring in ϕ is regarded as denoting $\mathsf{P}_{\mathfrak{A}}$,* (ii) *$\eta$ is interpreted as $\varepsilon_{\mathfrak{B}}$ and $=$ as the identity relation,* (iii) *the variables $\mathsf{v}_0, \ldots, \mathsf{v}_{n-1}$ are regarded (when occurring freely) as denoting $a_0, \ldots, a_{n-1}$ respectively, and* (iv) *for each ordinal α, the bound variables of ϕ of type α are regarded as ranging over the set $T_{\mathfrak{B}}(\alpha)$. As before, ϕ is said to be* true *in $\mathfrak{A}$ if ϕ contains no free variables and ϕ is satisfied in $\mathfrak{A}$ by Λ.*

Definitions 1 and 2 overlap in their range of applicability, but there is no danger of ambiguity. If ϕ is both a second-order formula and a higher-order formula, then all variables occurring in ϕ are of type 0, and ϕ is satisfied (or true) in the sense of Definition 1 if and only if ϕ is satisfied (or true) in the sense of Definition 2.

By a *formula* is understood either a second-order formula or a higher-order formula, and by a *sentence* a formula in which no variable is free. Two formulas ϕ and ψ with the same free variables are said to be *equivalent* if, for any model $\mathfrak{A}$ such that the language of $\mathfrak{A}$ contains all predicate constants occurring in ϕ or ψ, and any finite sequence a of elements of $U_{\mathfrak{A}}$ having as many places as ϕ and ψ have free variables, ϕ is satisfied in $\mathfrak{A}$ by a if and only if ψ is satisfied in $\mathfrak{A}$ by a.

To each second-order formula ϕ containing no predicate variables of more than one place, there corresponds in a natural way a higher-order formula ϕ'; we obtain ϕ' by replacing each subformula Pu, where P is a 1-place predicate variable and u an individual variable, by $\mathsf{u}\eta\mathsf{P}$. It can then be shown that ϕ is equivalent to ϕ'. Thus the decision to employ cumulative types has no effect on the truth of second-order formulas.

Lemma 1. *If $\mathfrak{A}$ and $\mathfrak{B}$ are isomorphic models and ϕ is a sentence containing no predicate constants beyond those in the language of $\mathfrak{A}$, then ϕ is true in $\mathfrak{A}$ if and only if ϕ is true in $\mathfrak{B}$.*

A direct generalization of second-order logic would require the introduction of predicate variables not only of various types but also of various numbers of places. The complications of such a course do not, however, seem worthwhile. What can be expressed with predicate variables of several places can be equivalently expressed with 1-place predicate variables, provided that one is willing to ascend in type.

By LT, or the class of *logical truths*, is understood the class of sentences ϕ such that ϕ is true in every model whose language contains all predicate constants occurring in ϕ.

The higher-order formulas clearly form a proper class; to change this we must restrict both the predicate constants and the types of variables that may occur. Let L be a set of predicate constants and Γ a set of ordinals. (In particular, Γ may be an ordinal.) Then by Γ-Fm_L, or the set of *Γ-order formulas* of L, is understood the set of higher-order formulas ϕ such that all variables occurring in ϕ have types in Γ and all predicate constants occurring in ϕ are in L; Γ–St_L, or the set of *Γ-order sentences* of L, is the set of Γ-order formulas of L having no free variables.

By II-Fm_L and II-St_L we understand respectively the set of second-order formulas all of whose predicate constants are in L and the set of ϕ in II-Fm_L such that ϕ has no free variables. By ESF_L, or the set of *existential second-order formulas* of L, is understood the set of ϕ in II-Fm_L such that ϕ has the form $\bigvee \mathsf{P}\psi$, where P is a 1-place predicate variable and no predicate variable is bound in ψ; USF_L is the set of negations of formulas in ESF_L.

Let ϕ be a sentence and L the set of predicate constants occurring in ϕ. By the *spectrum* of ϕ we understand the class of isomorphism types α such that ϕ is true in some L-model $\mathfrak{A}$ for which $\tau\mathfrak{A}=\alpha$. By inf (ϕ) and sup (ϕ) we understand the greatest lower bound and least upper bound respectively of the spectrum of ϕ with respect to $\leqslant$, provided that such bounds exist and are unique; otherwise inf (ϕ) and sup (ϕ) are the ordinal 0.

Let Φ be a class of sentences. A class Γ of isomorphism types is said to be *Φ-describable* if there is a member ϕ of Φ such that Γ is the spectrum of ϕ [3].

We shall be interested in two specializations of these notions. First, let L be the empty set Λ. Then we can identify the isomorphism type of an L-model $\mathfrak{A}$ with the *cardinal* of $U_{\mathfrak{A}}$; the class of all L-types will then coincide with the class of cardinal numbers [4]. If ϕ is in II-St_L, then the spectrum of ϕ coincides with what Zykov calls the spectrum of ϕ in Zykov [53]. The relation $\leqslant$ in this case amounts to the usual less-than-or-equal relation between cardinals, and is therefore a reflexive well-ordering. Consequently, if ϕ is a sentence true in at least one L-model, then inf (ϕ) exists and is a member of the spectrum of ϕ; and if ϕ is a sentence containing no predicate constants and such that the spectrum of ϕ has an upper bound, then sup (ϕ) exists. If Φ is any *set* of sentences containing no predicate constants, then the least upper bound of the set of cardinals inf (ϕ) for $\phi \varepsilon \Phi$ exists and is called the *index* of Φ, and the least upper bound of the set of cardinals sup (ϕ) for $\phi \varepsilon \Phi$ also exists and is called the *Hanf number* of Φ. (These two notions were introduced by William Hanf in [62c].) If Γ is a set of ordinals, then a *Γ-describable*

[3] The notions introduced above would also have interest if reference to an individual sentence were replaced by reference to a set of sentences.

[4] Notice that if we had not decided to admit empty models, there would be no L-type corresponding to the cardinal 0.

class (or set) *of cardinals* is a class (or set) of cardinals which is (Γ-St_A)-describable, and a *Γ-describable cardinal* is a cardinal α such that $\{\alpha\}$ is a Γ-describable set of cardinals. Similarly, a II-*describable class* (or set) *of cardinals* is one which is (II-St_A)-describable, and a II-*describable cardinal* is a cardinal α such that $\{\alpha\}$ is a II-describable set of cardinals.

For a second specialization we may assume that $L=\{\mathsf{P}\}$ and that P is a 2-place predicate constant. Then we can identify the L-types with *relation types*. In particular, we may consider those L-models $\mathfrak{A}$ for which $\mathsf{P}_{\mathfrak{A}}$ is a strict (or irreflexive) well-ordering (a condition equivalent to the truth in $\mathfrak{A}$ of the sentence $\Lambda \mathsf{x} \neg \mathsf{Pxx} \wedge \Lambda\mathsf{x}\Lambda\mathsf{y}\Lambda\mathsf{z}\ [\mathsf{Pxy} \wedge \mathsf{Pyz} \rightarrow \mathsf{Pxz}] \wedge \wedge \Lambda\mathsf{x}\Lambda\mathsf{y}\ [\mathsf{Pxy} \vee \mathsf{x}=\mathsf{y} \vee \mathsf{Pyx}] \wedge \Lambda\mathsf{Q}\ [\mathsf{V}\mathsf{x}\mathsf{x}\eta\mathsf{Q} \rightarrow \mathsf{V}\mathsf{x}(\mathsf{x}\eta\mathsf{Q} \wedge \neg\ \mathsf{V}\mathsf{y}\ [\mathsf{y}\eta\mathsf{Q} \wedge \mathsf{Pyx}])]$, where $\mathsf{x}, \mathsf{y}, \mathsf{z}$ are distinct individual variables and Q a variable of type 1); the type $\tau\mathfrak{A}$ of such a model may be identified with the *ordinal number* of the relational system $\langle U_{\mathfrak{A}}, \mathsf{P}_{\mathfrak{A}}\rangle$. The relation $\leqslant$ between types then amounts to the usual less-than-or-equal relation between ordinals, and is of course again a reflexive well-ordering. If Γ is a set of ordinals, then a *Γ-describable class* (or set) *of ordinals* is a class (or set) of ordinals which is (Γ-$\mathrm{St}_{\{\mathsf{P}\}}$)-describable for some 2-place predicate constant P, and a *Γ-describable ordinal* is an ordinal α such that $\{\alpha\}$ is a Γ-describable set of ordinals. We can characterize in an analogous way a II-*describable class of ordinals* and a II-*describable ordinal.*

The following lemma is due to David Kaplan.

Lemma 2. *If L is a set of predicate constants and ϕ is in* II-St_L, *then there exists ψ in* II-St_L *such that* (1) *ψ contains no predicate variables of more than two places and* (2) *for every model $\mathfrak{A}$, ψ is true in $\mathfrak{A}$ if and only if $U_{\mathfrak{A}}$ is infinite and ϕ is true in $\mathfrak{A}$.*

Most of our results will have to do with Γ-order logic, where Γ is a set of ω-describable ordinals. Various wider notions of a describable ordinal could replace ω-describability, but at the expense of complicating the constructions. The following lemma, which follows easily from Lemma 2, shows that ω-describability is at least sufficient to comprehend II-describability.

Lemma 3. (a) *Every* II-*describable cardinal or ordinal is a* 4-*describable cardinal or ordinal respectively.* (b) *Every* II-*describable class of infinite cardinals or ordinals is a* 4-*describable class of cardinals or ordinals respectively.*

2. Definition of the reduction. If L is a set of predicate constants and Γ a set of ordinals, then the set of *expressions corresponding to L and Γ* consists of (1) those second-order expressions in which no predicate constants occur beyond those in L, together with (2) those higher-order expressions in which no predicate constants occur beyond those in L and no variables occur beyond those having types in Γ. We say that v is a *new-variable function* for L and Γ if (1) v is a 3-place function, (2) v is defined for all triples $\langle k, n, \phi\rangle$ such that k, n are finite ordinals and ϕ is an expression corresponding to L and Γ, (3) for every finite ordinal n, every expression ϕ corresponding to L and Γ, and every u, u is a variable of type n not occurring in ϕ if and only if u is $v(k, n, \phi)$ for some finite ordinal k, and (4) whenever $\langle k, n, \phi\rangle$, $\langle k', n, \phi\rangle$ are triples for which v is defined, if $v(k, n, \phi)$ is $v(k', n, \phi)$, then k is k'. We may regard $v(k, n, \phi)$ as the kth variable of type n not occurring in ϕ (in the order determined by v).

Suppose that ψ is a function from variables to formulas. Then the *relativization* of ϕ by ψ, or $\phi^{(\psi)}$, is defined by the following recursion for any higher-order formula ϕ whose variables are all in the domain of ψ. (1) If P is an $(n+1)$-place predicate constant, $\mathsf{u}_0, \ldots, \mathsf{u}_n$ are individual variables, and u, v are variables of arbitrary types, then $(\mathsf{P}\mathsf{u}_0 \ldots \mathsf{u}_n)^{(\psi)}$ is $\mathsf{P}\mathsf{u}_0 \ldots \mathsf{u}_n$, $(\mathsf{u}\eta\mathsf{v})^{(\psi)}$ is $\mathsf{u}\eta\mathsf{v}$, and $(\mathsf{u}=\mathsf{v})^{(\psi)}$ is $\mathsf{u}=\mathsf{v}$. (2) $(\neg\, \phi)^{(\psi)}$ is $\neg\, \phi^{(\psi)}$; $(\phi \to \chi)^{(\psi)}$ is $(\phi^{(\psi)} \to \chi^{(\psi)})$; and similarly for the other sentential connectives. (3) $(\wedge \mathsf{u}\phi)^{(\psi)}$ is $\wedge \mathsf{u}(\psi(\mathsf{u}) \to \phi^{(\psi)})$, and $(\vee \mathsf{u}\phi)^{(\psi)}$ is $\vee \mathsf{u}(\psi(\mathsf{u}) \wedge \phi^{(\psi)})$.

For the remainder of this section assume that ε is a 2-place predicate constant, Σ a 1-place predicate constant, L a set of predicate constants, Γ a set of ordinals, and v a new-variable function for $L \cup \{\varepsilon, \Sigma\}$ and $\Gamma \cup \omega$.

If g is a biunique function from variables to individual variables and ϕ is a higher-order formula all of whose variables are in the domain of g, then the *replacement* in ϕ based on g and ε, or $\phi_{g,\varepsilon}$, is defined by the following recursion. (1) If P is an $(n+1)$-place predicate constant, $\mathsf{u}_0, \ldots, \mathsf{u}_n$ are individual variables, and u, v are variables of arbitrary type, then $(\mathsf{P}\mathsf{u}_0 \ldots \mathsf{u}_n)_{g,\varepsilon}$ is $\mathsf{P}g(\mathsf{u}_0) \ldots g(\mathsf{u}_n)$, $(\mathsf{u}=\mathsf{v})_{g,\varepsilon}$ is $g(\mathsf{u})=g(\mathsf{v})$, and $(\mathsf{u}\eta\mathsf{v})_{g,\varepsilon}$ is $g(\mathsf{u})\, \varepsilon\, g(\mathsf{v})$. (2) $(\neg\, \phi)_{g,\varepsilon}$ is $\neg\, (\phi_{g,\varepsilon})$; $(\phi \to \psi)_{g,\varepsilon}$ is $(\phi_{g,\varepsilon} \to \psi_{g,\varepsilon})$; and similarly for the other sentential connectives. (3) $(\wedge \mathsf{u}\phi)_{g,\varepsilon}$ is $\wedge g(\mathsf{u})\phi_{g,\varepsilon}$, and $(\vee \mathsf{u}\phi)_{g,\varepsilon}$ is $\vee g(\mathsf{u})\phi_{g,\varepsilon}$.

We understand J to be the *Cantor pairing function*, which in a biunique and recursive way assigns to every pair of finite ordinals a finite ordinal.

If ϕ is in ω-$\mathrm{St}_{\{\varepsilon\}}$ and a is an individual variable, then $\mathrm{OT}_{\phi,\varepsilon,v}(\mathsf{a})$ is to be the formula $(\phi_{g,\varepsilon})^{(\psi)}$, where g is that function whose domain is the set of variables of finite type and such that for all finite ordinals k and n, $g(v(k, n, \Lambda))$ is $v(J(k, n), 0, \mathsf{a})$, and ψ is that function whose domain is the set of variables of type 0 and such that whenever u is a variable of type 0, $\psi(g(\mathsf{u}))$ is the formula $g(\mathsf{u})\varepsilon\mathsf{a}$, and whenever n is a finite ordinal and u is a variable of type $n+1$, $\psi(g(\mathsf{u}))$ is the formula $\bigvee g(\mathsf{w})\ [\bigwedge g(\mathsf{t})\ [g(\mathsf{t})\varepsilon g(\mathsf{w}) \leftrightarrow \psi(g(\mathsf{t}))] \wedge \bigwedge g(\mathsf{t})\ [g(\mathsf{t})\varepsilon g(\mathsf{u}) \rightarrow g(\mathsf{t})\varepsilon g(\mathsf{w})]]$ (where t is $v(0, n, \Lambda)$ and w is $v(0, n+1, \mathsf{u})$). We shall be interested in $\mathrm{OT}_{\phi,\varepsilon,v}(\mathsf{a})$ only when ϕ is a sentence whose spectrum is $\{\alpha\}$, for some ordinal α; and in this case $\mathrm{OT}_{\phi,\varepsilon,v}(\mathsf{a})$ may be read 'a is a set having α as its order type.'

If again ϕ is in ω-$\mathrm{St}_{\{\varepsilon\}}$ and a is an individual variable, then $\mathrm{T}_{\phi,\varepsilon,\Sigma,v}(\mathsf{a})$ is to be the formula $\bigvee\mathsf{b}\ [\Sigma\mathsf{b} \wedge \bigwedge\mathsf{x}(\mathsf{x}\varepsilon\mathsf{b} \leftrightarrow \bigwedge\mathsf{k}\ [\bigwedge\mathsf{m}\bigwedge\mathsf{n}(\bigwedge\mathsf{y}\ [\mathsf{y}\varepsilon\mathsf{m} \rightarrow \mathsf{y}\varepsilon\mathsf{k}] \wedge \wedge \bigwedge\mathsf{y}\ [\mathsf{y}\varepsilon\mathsf{n} \leftrightarrow \neg\ \Sigma\mathsf{y} \vee \bigvee\mathsf{x}(\mathsf{x}\varepsilon\mathsf{m} \wedge \bigwedge\mathsf{z}\ [\mathsf{z}\varepsilon\mathsf{y} \rightarrow \mathsf{z}\varepsilon\mathsf{x}])] \rightarrow \mathsf{n}\varepsilon\mathsf{k} \vee \bigwedge\mathsf{y}\ [\mathsf{y}\varepsilon\mathsf{b} \rightarrow \mathsf{y}\varepsilon\mathsf{n}]) \rightarrow \rightarrow \mathsf{x}\varepsilon\mathsf{k}]) \wedge \mathrm{OT}_{\phi,\varepsilon,v}(\mathsf{b}) \wedge \Sigma\mathsf{a} \wedge \bigwedge\mathsf{y}\ [\mathsf{y}\varepsilon\mathsf{a} \leftrightarrow \neg\ \Sigma\mathsf{y} \vee \bigvee\mathsf{x}(\mathsf{x}\varepsilon\mathsf{b} \wedge \bigwedge\mathsf{z}\ [\mathsf{z}\varepsilon\mathsf{y} \rightarrow \mathsf{z}\varepsilon\mathsf{x}])]]$, where $\mathsf{b}, \mathsf{k}, \mathsf{m}, \mathsf{n}, \mathsf{x}, \mathsf{y}, \mathsf{z}$ are respectively $v(0, 0, \mathsf{a}), \ldots, v(6, 0, \mathsf{a})$. If the spectrum of ϕ is $\{\alpha\}$ and α is an ordinal, then $\mathrm{T}_{\phi,\varepsilon,\Sigma,v}(\mathsf{a})$ may be read 'a is the αth cumulative type'.

If a is an individual variable and n a finite ordinal, then $\mathrm{Dom}_{n,\varepsilon,v}(\mathsf{a})$ is to be the formula $\mathsf{a}=\mathsf{a}$ if n is 0, and the formula $\bigvee\mathsf{b}_1 \ldots \bigvee\mathsf{b}_n\ [\mathsf{a}\varepsilon\mathsf{b}_1 \wedge \ldots \ldots \wedge \mathsf{b}_{n-1}\varepsilon\mathsf{b}_n]$, where $\mathsf{b}_1, \ldots, \mathsf{b}_n$ are respectively $v(0, 0, \mathsf{a}), \ldots, v(n-1, 0, \mathsf{a})$, if $n>0$. We may read $\mathrm{Dom}_{n,\varepsilon,v}(\mathsf{a})$ as 'a is n-dominated'.

Suppose that g is a biunique function from variables to individual variables, ϕ is a sentence in Γ-St_L all of whose variables are in the domain of g, n is a finite ordinal, $\mathsf{u}_0, \ldots, \mathsf{u}_k$ are all the variables occurring in ϕ, listed in order of first occurrence, and d is a function whose domain contains $\mathsf{u}_0, \ldots, \mathsf{u}_k$ and such that $d(\mathsf{u}_0), \ldots, d(\mathsf{u}_k)$ are in $(n+2)$-$\mathrm{St}_{\{\varepsilon\}}$. Then the corresponding *second-order reduction* of ϕ, or $\mathrm{Red}_{g,d,\varepsilon,\Sigma,n,v}(\phi)$, is the sentence $\mathrm{RFS}_{\varepsilon,\Sigma,\mathsf{b}_0,\ldots,\mathsf{b}_6,\mathsf{P}} \wedge \bigvee\mathsf{a}_0 \ldots \bigvee\mathsf{a}_k[\mathrm{T}_{d(\mathsf{u}_0),\varepsilon,\Sigma,v}(\mathsf{a}_0) \wedge \ldots \wedge \mathrm{T}_{d(\mathsf{u}_k),\varepsilon,\Sigma,v}(\mathsf{a}_k) \wedge \mathrm{Dom}_{n,\varepsilon,v}(\mathsf{a}_0) \wedge \ldots \wedge \mathrm{Dom}_{n,\varepsilon,v}(\mathsf{a}_k) \wedge [\neg\,\mathrm{Dom}_{n+1,\varepsilon,v}(\mathsf{a}_0) \vee \ldots \vee \neg\,\mathrm{Dom}_{n+1,\varepsilon,v}(\mathsf{a}_k)] \wedge (\phi_{g,\varepsilon})^{(\psi)}]$, where $\mathsf{a}_0, \ldots, \mathsf{a}_k, \mathsf{b}_0, \ldots, \mathsf{b}_6, \mathsf{P}$ are respectively $v(0, 0, \phi_{g,\varepsilon}), \ldots \ldots, v(k, 0, \phi_{g,\varepsilon}), v(0, 0, \Lambda), \ldots, v(6, 0, \Lambda), v(0, 1, \Lambda)$, and ψ is that function whose domain is $\{g(\mathsf{u}_0), \ldots, g(\mathsf{u}_k)\}$ and such that, for $i \leqslant k$, $f(g(\mathsf{u}_i))$ is the formula $g(\mathsf{u}_i)\varepsilon\mathsf{a}_i$.

3. Spectrum problems.

Definition 3. *If α is a cardinal and β an ordinal, then $\beth_{\alpha,\beta}$ is to be a cardinal determined in accordance with the following recursive clauses:*

$\beth_{\alpha,0}$ *is* α; $\beth_{\alpha,\beta+1}$ *is* $\alpha+2^{\beth_{\alpha,\beta}}$; *if* β *is a limit ordinal, then* $\beth_{\alpha,\beta}$ *is the least upper bound of the cardinals* $\beth_{\alpha,\gamma}$ *for* $\gamma<\beta$.

Theorem 1. *Let n be a finite ordinal, Γ be any set of $(n+2)$-describable ordinals, ϕ be in Γ-St_Λ, and β be the maximum of the types of variables occurring in ϕ. Then there exists a sentence ψ in Π-St_Λ such that the spectrum of ψ is the class of cardinals $\beth_{\alpha,\beta+1+n}$ for which α is in the spectrum of ϕ; ψ may, moreover, be chosen so as to have the form* $\mathsf{VPVQ}\delta$, *where* P *is a 2-place predicate variable,* Q *is a 1-place predicate variable, and δ is in* USF_Λ.

Proof. Let ε be a 2-place predicate constant and Σ a 1-place predicate constant. Let g be a biunique function whose domain is the set of variables occurring in ϕ and whose values are individual variables. Clearly there is a function d with the same domain such that whenever u is a variable of type α occurring in ϕ, $d(\mathsf{u})$ is in $(n+2)$-$\mathrm{St}_{\{\varepsilon\}}$ and the spectrum of $d(\mathsf{u})$ is $\{\alpha\}$. Let v be a new-variable function for $\{\varepsilon, \Sigma\}$ and $\Gamma \cup \omega$, let P be a 2-place predicate variable, let Q be a 1-place predicate variable not occurring in $\mathrm{Red}_{g,d,\varepsilon,\Sigma,n,v}(\phi)$, and let χ be the result of replacing ε by P and Σ by Q in $\mathrm{Red}_{g,d,\varepsilon,\Sigma,n,v}(\phi)$. It is easily seen that $\mathsf{VPVQ}\chi$ is equivalent to some sentence of the form $\mathsf{VPVQ}\delta$, where δ is in USF_Λ. We may then take ψ to be $\mathsf{VPVQ}\delta$.

It is clear that if ϕ is in Π-St_L (where L is any set of predicate constants), then there exists a sentence ϕ' in ω-St_L such that ϕ is equivalent to ϕ'. Thus, as a special case of Theorem 1, we may let n be 0 and replace the assumption that ϕ is in Γ-St_Λ by the assumption that ϕ is in Π-St_Λ; the resulting assertion corresponds closely to Theorem III of Zykov [53], which, however, contains slight advantages that could not be expected in a general theorem.

If Γ is any set of cardinals, understand by $<[\Gamma]$ the restriction of $<$ to Γ, that is, the set of ordered pairs $\langle\alpha, \beta\rangle$ such that α, β are in Γ and $\alpha<\beta$. The following is a simple consequence of Lemma 3 and Theorem 1.

Theorem 2. *Assume the Generalized Continuum Hypothesis. If n is a finite ordinal, $4\leqslant n$, Γ is the set of n-describable cardinals, and Δ is the set of Π-describable cardinals, then $<[\Gamma]$ and $<[\Delta]$ have the same order type.*

This consequence was pointed out by Barry Kurtzman, who has obtained an independent proof not involving the Continuum Hypothesis.

Theorem 3. *Suppose that* Γ *is a set of* ω*-describable ordinals,* $4 \subseteq \Gamma$, α *and* α' *are respectively the index and the Hanf number of* Γ-St_A, β *and* β' *are respectively the index and the Hanf number of* II-St_A, *and* γ *and* γ' *are respectively the index and the Hanf number of the set of sentences in* II-St_A *having the form* $\mathsf{V}\mathsf{P}\mathsf{V}\mathsf{Q}\phi$ *(where* P *is a 2-place predicate variable,* Q *is a 1-place predicate variable, and* ϕ *is in* USF_A*). Then*

$$\alpha = \beta = \gamma < \alpha' = \beta' = \gamma'.$$

The equalities in this theorem follow immediately from Lemma 2 and Theorem 1. The fact that $\gamma < \alpha'$ requires a separate proof, which will not, however, be given here.

If Γ satisfies the hypothesis of Theorem 3, then in view of Theorems 2 and 3 and Lemma 3(a) we might suppose that the Γ-describable cardinals and the II-describable cardinals coincide. It follows from the next theorem, which will not be proved here, that this supposition would be false.

Theorem 4. *Let* n *be a finite ordinal which is at least* 3, *and let* α *be the first cardinal which is not* n*-describable. Then* α *is* $(n+1)$*-describable. If, moreover,* $n \geqslant 4$, *then* α *is not* II*-describable.*

4. Logical truth. If L is a countable (that is, at most denumerable) set of predicate constants and Γ a countable set of ordinals, then nr is said to be a *Gödel-numbering corresponding to* L *and* Γ if (1) nr is a biunique function whose values are finite ordinals, (2) the domain of nr is the set of expressions corresponding to L and Γ, and (3) the following sets and relations are recursive: the set of numbers $nr(\mathsf{P})$ such that P is a predicate constant in L, the set of numbers $nr(\mathsf{P})$ such that P is a predicate variable, the set of numbers $nr(\mathsf{u})$ such that u is a variable having type in Γ, the range of nr, the set of pairs $\langle nr(\mathsf{P}), n\rangle$ such that n is in ω and P is an n-place predicate constant in L, the set of pairs $\langle nr(\mathsf{P}), n\rangle$ such that n is in ω and P is an n-place predicate variable, the set of pairs $\langle nr(\mathsf{u}), n\rangle$ such that n is in $\omega \cap \Gamma$ and u is a variable of type n, and the set of triples $\langle nr(\phi), nr(\psi), nr(\chi)\rangle$ such that ϕ, ψ are in the domain of nr and χ is the concatenation of ϕ and ψ.

Assume in what follows that L is a countable set of predicate constants, Γ is a countable set of ordinals, ε is a 2-place predicate constant not in L, Σ is a 1-place predicate constant not in L, nr is a Gödel-numbering corresponding to $L \cup \{\varepsilon, \Sigma\}$ and $\Gamma \cup \omega$, and S, P are distinct 3-place predicate constants.

If R is a set of, or relation between, expressions corresponding to $L \cup \{\varepsilon, \Sigma\}$ and $\Gamma \cup \omega$, then R is said to be *nr-recursive* if the image of R under nr (which will be a set of, or relation between, finite ordinals) is recursive. An n-place function is regarded as a special case of an $(n+1)$-place relation, and therefore no special definition of nr-recursivity for functions is required.

Theorem 5. *Suppose that an nr-recursive function d exists whose domain is the set of variables having types in Γ, and such that whenever* u *is a variable of type α and α is in Γ, $d(u)$ is in ω-$\mathrm{St}_{\{\varepsilon\}}$ and the spectrum of $d(\mathsf{u})$ is $\{\alpha\}$. Then* $\mathrm{LT} \cap \Gamma$-St_L *is recursively reducible to* $\mathrm{LT} \cap \mathrm{ESF}_{L\cup\{\varepsilon,\Sigma\}}$; *indeed, there exists an nr-recursive function f with domain Γ-St_L, with range included in* $\mathrm{ESF}_{L\cup\{\varepsilon,\Sigma\}}$, *and such that whenever ϕ is in Γ-St_L, we have: ϕ is in* $\mathrm{LT} \cap \Gamma$-St_L *if and only if $f(\phi)$ is in* $\mathrm{LT} \cap \mathrm{ESF}_{L\cup\{\varepsilon,\Sigma\}}$.

Proof. Clearly there is an nr-recursive new-variable function v for $L \cup \{\varepsilon, \Sigma\}$ and $\Gamma \cup \omega$, and there is a biunique nr-recursive function g whose domain is the set of variables having types in Γ and whose values are individual variables. Let h be that function with domain Γ-St_L such that for each ϕ in Γ-St_L, $h(\phi)$ is the least finite ordinal n such that whenever u is a variable having type in Γ and occurring in ϕ, $d(\mathsf{u})$ is in $(n+2)$-$\mathrm{St}_{\{\varepsilon\}}$. Then clearly h is nr-recursive. Let t be that function with domain Γ-St_L such that for each ϕ in Γ-St_L, $t(\phi)$ is the sentence $\neg\, \mathrm{Red}_{g,d,\varepsilon,\Sigma,h(\Phi),v}\,(\neg\, \phi)$. It is easily seen that t is recursive. The desired function f can now be obtained from t; intuitively speaking, $f(\phi)$ is to be the result of moving the second-order quantifier to the front in $t(\phi)$.

The special case of Theorem 5 in which Γ is ω is essentially Theorem III of Hintikka [55b]. As an immediate consequence of Theorem 5 we have the following.

Theorem 6. *If Γ is a finite set of ω-describable ordinals, then the conclusion of Theorem 5 holds.*

By $\mathfrak{N}_{\mathsf{S},\mathsf{P}}$, or the *numerical model* corresponding to S and P, is understood that $\{\mathsf{S}, \mathsf{P}\}$-model $\mathfrak{A}$ such that (1) $U_{\mathfrak{A}}$ is ω, (2) $\mathsf{S}_{\mathfrak{A}}$ is the set of triples $\langle m, n, p\rangle$ for which m, n are finite ordinals and p is $m+n$, and (3) $\mathsf{P}_{\mathfrak{A}}$ is the set of triples $\langle m, n, p\rangle$ for which m, n are finite ordinals and p is $m\cdot n$. If $\mathfrak{A}$ is any model and B a subset of $\mathfrak{A}$, then ϕ is said to *define B* in $\mathfrak{A}$ if (1) ϕ is a formula with exactly one free individual variable and no free variables which are not individual variables, (2) all predicate constants occurring in ϕ are in the language of $\mathfrak{A}$, and (3) for all x in $U_{\mathfrak{A}}$, x is in B if and only if the one-place sequence $\langle x \rangle$ satisfies ϕ in $\mathfrak{A}$. If in addition Δ is any set of ordinals, then B is said to be *Δ-definable* in $\mathfrak{A}$ if there exists ϕ in $\Delta\text{-Fm}_L$, where L is the language of $\mathfrak{A}$, such that ϕ defines B in $\mathfrak{A}$. By $f\star(A)$ is understood the set of objects $f(x)$ such that x is in A.

It is a consequence of Gödel's completeness theorem that $nr\star(\text{LT} \cap \cap\, 1\text{-Fm}_L)$ and $nr\star(\text{LT} \cap \text{USF}_L)$ are 1-definable in $\mathfrak{N}_{\mathsf{S},\mathsf{P}}$. It is rather easily seen that the same cannot be said for the set of Gödel numbers of second-order logical truths. It is natural, however, to ask whether this set occurs in some natural extension of the Kleene arithmetical hierarchy, for instance, whether it is 2-definable or ω-definable in $\mathfrak{N}_{\mathsf{S},\mathsf{P}}$. The next theorem, which was derived from Theorem 6 by Vaught and me, gives a negative answer; indeed, we have the stronger result that in a 'natural' higher-order language not even the set of Gödel numbers of *existential* second-order logical truths is definable. First a lemma is required which asserts the existence of self-referential sentences.

Lemma 4. *If 0 is in Γ, f is an nr-recursive function whose domain includes $\Gamma\text{-St}_{\{\mathsf{S},\mathsf{P}\}}$, ϕ is in $\Gamma\text{-Fm}_{\{\mathsf{S},\mathsf{P}\}}$, ϕ has exactly one free variable, and that is an individual variable, then there exists ψ in $\Gamma\text{-St}_{\{\mathsf{S},\mathsf{P}\}}$ such that ψ is true in $\mathfrak{N}_{\mathsf{S},\mathsf{P}}$ if and only if $\langle nr(f(\psi))\rangle$ satisfies ϕ in $\mathfrak{N}_{\mathsf{S},\mathsf{P}}$.*

Theorem 7. *If all members of Γ are ω-describable ordinals and L contains S and P, then $nr\star(\text{LT} \cap \text{ESF}_{L \cup \{\varepsilon, \Sigma\}})$ is not Γ-definable in $\mathfrak{N}_{\mathsf{S},\mathsf{P}}$.*

Proof. Assume the hypothesis and that ϕ is a formula in $\Gamma\text{-Fm}_{\{\mathsf{S},\mathsf{P}\}}$ which defines $nr\star(\text{LT} \cap \text{ESF}_{L \cup \{\varepsilon, \Sigma\}})$ in $\mathfrak{N}_{\mathsf{S},\mathsf{P}}$. Let Δ be the set consisting of the ordinals 0 and 1, together with the types of variables occurring in ϕ. By Theorem 6, there exists an nr-recursive function f with domain $\Delta\text{-St}_{\{\mathsf{S},\mathsf{P}\}}$ such that (1) the range of f is included in $\text{ESF}_{\{\mathsf{S},\mathsf{P},\varepsilon,\Sigma\}}$ and (2) for all ψ in $\Delta\text{-St}_{\{\mathsf{S},\mathsf{P}\}}$, ψ is in $\text{LT} \cap \Delta\text{-St}_{\{\mathsf{S},\mathsf{P}\}}$ if and only if $f(\psi)$ is in $\text{LT} \cap \text{ESF}_{\{\mathsf{S},\mathsf{P},\varepsilon,\Sigma\}}$. It is well known that there exists a sentence χ in $2\text{-St}_{\{\mathsf{S},\mathsf{P}\}}$ such that for every $\{\mathsf{S}, \mathsf{P}\}$-model $\mathfrak{A}$, χ is true in $\mathfrak{A}$ if and only if $\mathfrak{A}$ is isomorphic to $\mathfrak{N}_{\mathsf{S},\mathsf{P}}$. We now employ the Paradox of the Liar. By Lemma 4, there exists ψ in $\Delta\text{-St}_{\{\mathsf{S},\mathsf{P}\}}$ such that ψ is true in $\mathfrak{N}_{\mathsf{S},\mathsf{P}}$ if and only if $\langle nr\,(f(\chi \rightarrow \psi))\rangle$ satisfies $\neg\,\phi$ in $\mathfrak{N}_{\mathsf{S},\mathsf{P}}$. It follows by (1), (2), and Lemma 1 that ψ is true in $\mathfrak{N}_{\mathsf{S},\mathsf{P}}$ if and only if ψ is not true in $\mathfrak{N}_{\mathsf{S},\mathsf{P}}$, and the theorem is proved.

In contrast with Theorem 7 we have the following positive definability result, whose proof, though simple, will not be given here.

Theorem 8. *If* α *is the index of* II-St$_A$, *then* $nr\star$(LT $\cap$ II-St$_L$) *is* $(\alpha+1)$*-definable in* $\mathfrak{N}_{\mathsf{S,P}}$.

I do not know whether $\alpha+1$ is the least ordinal satisfying the conclusion of Theorem 8.